

10 Non-Hierarchical Clustering

MA2003B Application of Multivariate Methods
in Data Science

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There are many methods for non-hierarchical clustering, but most of them can be classified as **partitioning methods**, **density-based methods** or **model-based methods**.

Partitioning Methods

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The most common partitioning methods are:

- **k-means clustering:** This method aims to partition the data into k clusters such that the sum of squared distances between each point and the centroid of its assigned cluster is minimized. The algorithm iteratively assigns points to the nearest centroid and updates the centroids until convergence.

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- **CLARA (Clustering LARge Applications)**: An extension of k-medoids that is designed to handle large datasets by sampling a subset of the data and applying the k-medoids algorithm to this sample.
- **PAM (Partitioning Around Medoids)**: A specific implementation of the k-medoids algorithm that uses a greedy approach to find the medoids and assign points to clusters.

Density-Based Methods

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The most common density-based methods are:

- **DBSCAN (Density-Based Spatial Clustering of Applications with Noise)**: This method defines clusters as areas of high density separated by areas of low density. It requires two parameters: the radius ϵ that defines the neighborhood around a point, and the minimum number of points required to form a dense region. Points that do not belong to any cluster are considered noise.

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- **OPTICS (Ordering Points To Identify the Clustering Structure)**: An extension of DBSCAN that creates an ordering of the points based on their density, allowing for the identification of clusters at different density levels.
- **Mean Shift**: This method iteratively shifts each point towards the mean of the points in its neighborhood, effectively finding the modes of the data distribution. Clusters are formed around these modes.

Model-Based Methods

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The most common model-based methods are:

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- **Bayesian Clustering**: This method uses Bayesian inference to estimate the parameters of the underlying distributions and to assign points to clusters. It can incorporate prior knowledge about the data and can handle uncertainty in the clustering process.
- **Latent Class Analysis (LCA)**: This method is used for categorical data and assumes that the data is generated from a mixture of latent classes. It estimates the parameters of the classes and assigns points to classes based on their posterior probabilities.

Choosing the Right Method

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Here are some guidelines for choosing the right method:

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- If the data is categorical, Latent Class Analysis (LCA) may be the most appropriate method.
- If the dataset is large, consider using methods like CLARA that are designed to handle large datasets efficiently.

Evaluation of Clustering Results

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 - **Davies-Bouldin Index:** Evaluates the average similarity ratio of each cluster with its most similar cluster. Lower values indicate better clustering.
 - **Calinski-Harabasz Index:** Measures the ratio of between-cluster variance to within-cluster variance. Higher values indicate better-defined clusters.

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 - **Adjusted Rand Index (ARI):** Measures the similarity between the clustering results and the ground truth, adjusted for chance. Higher values indicate better agreement.
 - **Normalized Mutual Information (NMI):** Measures the amount of information shared between the clustering results and the ground truth. Higher values indicate better agreement.

- **Stability Analysis:** This involves assessing the robustness of the clustering results by applying the clustering algorithm to different subsets of the data or by adding noise to the data. Consistent clustering results across different runs indicate stable clusters.

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- **Visual Inspection:** Visualizing the clustering results using scatter plots, heatmaps, or dendrograms can provide insights into the structure of the clusters and help identify any potential issues with the clustering.

Example in R

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The methods we will compare are:

- **k-means clustering (a partitioning method)**
- **PAM (Partitioning Around Medoids, a k-medoids method)**
- **Gaussian Mixture Models (GMM, a model-based method)**
- **DBSCAN (a density-based method)**

Iris sepal-based clustering

